Lecture 1:
Introducing CS5702 - Modern Data

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Lecture Poll

Please use your mobile or laptop to go to:

pollev.com/mshepperd

We will need it in a few minutes. Thanks!

Welcome

This is a great time to study data science, statistics and artificial intelligence.

(The image was generated by DALL-E using the prompt "painting of very happy students studying data science in style of Lucian Freud".)



Agenda

1. Module overview
2. Teaching approach and resources
3. What is R and why do data scientists use it?
4. Where does generative AI fit?
5. Getting help
6. Extension question and study

1. Module overview

Meet the team

— Prof Martin Shepperd (module leader)
— Dr Ferdoos Hussein-Nezhad (co-module leader)
— Dr Valeriia Haberland (lecturer)
— Dima Alattal (GTA)
— Nehal Niloy (GTA)
— Duum Okpoechi (GTA)
— Suzy Zhu (GTA)

Lecture protocol

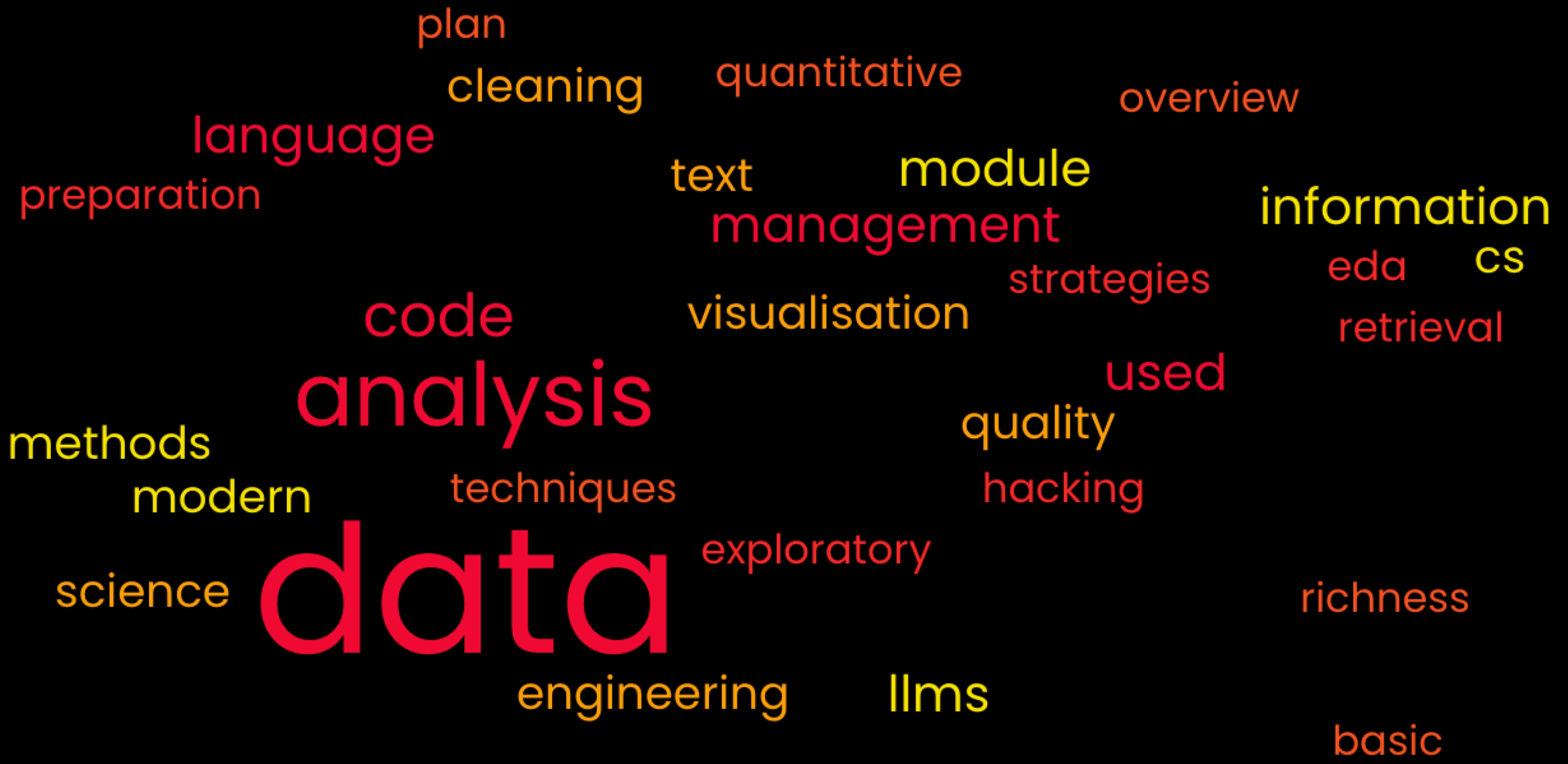
1. I will start at **1005 prompt**; please be ready
2. Be aware, lectures will be recorded
3. Feel free to ask **questions** as we go along or ...
4. ... ask during a question gap (usually 2-3 per lecture)
5. Be **considerate** of others (fellow students and me) and don't chat.
6. Thanks!

What is data?

Wisdom of crowds



Source: Kozalov and Valkov, https://


plan
cleaning
quantitative
overview
language
text
module
preparation
management
information
eda
cs
strategies
code
visualisation
retrieval
analysis
used
quality
methods
modern
techniques
hacking
data
exploratory
science
richness
engineering
llms
basic

What's Modern Data about?

To provide an introduction to **data management** and **exploration**. ... appreciation of the richness and availability of different data sources ... **techniques, methods and processes** for modern data analysis.

— Study Guide

Data science is ...

Data science is an **exciting** discipline that allows you to turn **raw** data into **understanding**, **insight**, and **knowledge**.

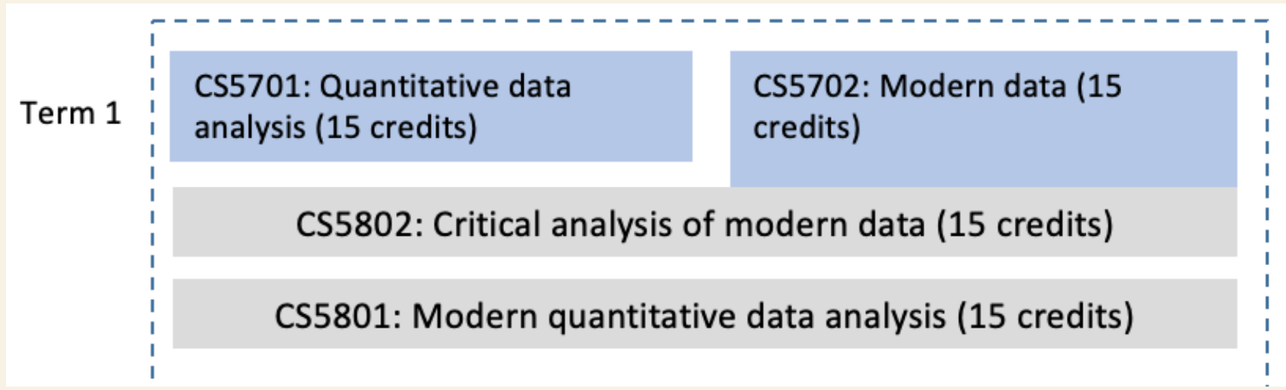
— Hadley Wickham and Garrett Grolemund

Opportunities

... there is more data and richer data available than ever before, coupled with more and more powerful analysis tools.

... incredible **opportunities** to collect, clean, merge, analyse and visualise data both for **good**, and sadly for less good purposes.

Module structure



Week structure

Date	Week	Lecture Topic	Lecturer	Lab Activity
23/09/2026	0	Induction material for ‘newbies’		
30/09/2026	1	Module overview, motivating examples, the R ecosystem	MS	What is data science? What can we do with R? Basic R and descriptive stats.
07/10/2026	2	The richness of data: structured and unstructured	MS	Analysing the module survey.
14/10/2026	3	Engineering or hacking? Readable code and reproducible data analysis	MS	Finding and importing data. Simple visualisation of time series, smoothing.
21/10/2026	4	Exploratory data analysis (EDA) and visualisation	FHN	Exploring data: more complex questions (salary and gender analysis)
28/10/2026	5	Data quality, cleaning and imputation	VH	Cleaning data set examples. Imputation with R packages.
04/11/2026	6	READING WEEK		
11/11/2026	7	Presenting data effectively	MS	Generative AI and your workflow.
18/11/2026	8	NLP and text processing	FHN	Text mining, word clouds.
25/11/2026	9	Sentiment analysis	FHN	Sentiment analysis in R.
02/12/2026	10	Data dimensionality reduction	FHN	Exploring High-Dimensional Data with PCA and Autoencoders in R
09/12/2026	11	Exam (CS5802) revision	MS	Course work support.

2. Teaching approach and resources

- Practical
- Lecture and lab
- Use R and RStudio
- Being organised
  - Decide on your personal to-do / checklist system **now**!
  - Keep up to date
  - Weekly checklists (as text files) can be grabbed from here

Learning Resources:

— Brightspace VLE

— The "Modern Data" interactive book

— Worksheets and linked files from Brightspace/GitHub

— Quizzes

— Reading list and references (also your own research)

3. What is R and why do data scientists use it?

R is an open, purpose-designed, highly-extensible, statistical and data analysis programming language.

R Advantages

- designed by statisticians
- powerful data handling, wrangling and storage capabilities
- flexible graphical facilities
- integrates with machine learning tools e.g., TensorFlow etc
- interactive dashboards
- large, open community
- easy integration with e.g., C, C++, Python
- widely used by researchers

PYPL Index (PopularitY of Programming Language)

Source: https://pypl.github.io/PYPL.html

Worldwide, Sept 2025 :

Rank	Change	Language	Share	1-year trend
1		Python	29.69 %	+0.2 %
2		Java	14.72 %	-0.7 %
3	↑	C/C++	9.27 %	+2.5 %
4	↓	JavaScript	6.79 %	-1.4 %
5	↑	R	5.26 %	+0.6 %
6	↓	C#	4.81 %	-1.8 %
7	↑↑↑↑	Objective-C	4.17 %	+1.7 %
8	↓	PHP	3.34 %	-0.8 %
9	↑	Rust	2.73 %	+0.2 %
10	↓	Swift	2.72 %	+0.0 %

R basics

As a **prerequisite** you should have:

completed the Getting Ready Chapter, in particular to have **installed** R and RStudio and run some simple R test examples.

(OR the equivalent Week0 on Brightspace)

R and variables

A variable is a named container for data and this data can be set, modified or referenced.

```
# This R code creates three different variables

numericVariable <- 10.3
charVariable <- "Hello world!"
logicVariable <- TRUE
```

R determines the **data type** from the value you assign.

Data type

In R, type describes the underlying object/value and determines how R treats it and what operations make sense.

It:
- which operations make sense
- how the data are interpreted
- what values can be represented

Simple data types in R

— numeric (floating point)

— integer

— character (text)

— logical (TRUE or FALSE)

Manipulating variables

```
# Initialise y (or overwrite if it already exists) to 5.3
y <- 5.3

# Multiply y by 13
y <- y * 13

# Display y
y
[1] 68.9
```

R and more complex data

I'm deliberately stopping the R teaching here.

This afternoon in the lab we'll start using this properly, including vectors and data frames.

4. Where does generative AI fit?

— LLMs[0] such as ChatGPT, Claude, etc can generate R code remarkably well

— but plausible-looking answers can still be wrong

— you need enough understanding to judge the answer

— so use it with caution. We will revisit and explore more about specifying and testing in Week 6.

[0] A large language model (LLM) is an AI system trained on vast amounts of text data to understand, generate and respond to human language in a coherent and contextually relevant way. For an introductory source see OpenAI's documentation on GPT models.

ChatGPT - write me an R program ...

... to produce a histogram

MA

Write a R program to produce a histogram

Sure, here's an example of an R program that generates a histogram using some sample data:

R Copy code

```r
# Sample data
data <- c(12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 5

# Create a histogram
hist(data,
     main = "Histogram Example",
     xlab = "Values",
     ylab = "Frequency",
     col = "skyblue",
     border = "black",
     breaks = 5,    # Number of bins
     xlim = c(10, 70)   # X-axis limits
)
```

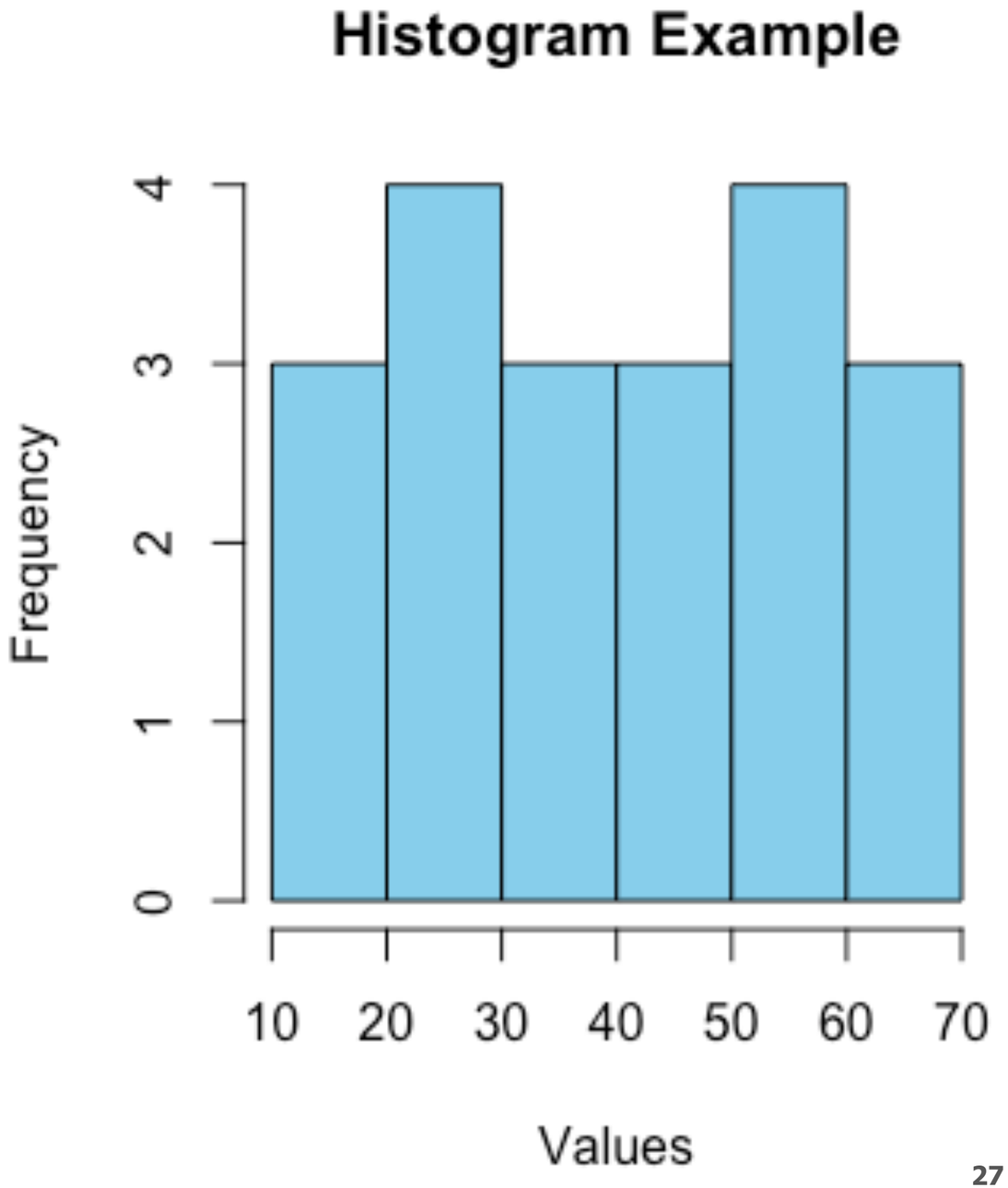
In this example, the `data` vector contains the sample data points. The `hist` function is used to create the histogram, and you can customize the appearance and behavior of the histogram using various parameters such as `main`, `xlab`, `ylab`, `col`, `border`, `breaks`, and `xlim`.

Make sure you have R installed on your computer and you can run this code in an R interpreter or script editor to see the histogram generated using the provided sample data. You can also replace the `data` vector with your own data to visualize a different dataset.

Regenerate

et voilà!

BTW you may need to paste the code into RStudio (or R) to execute it.



ChatGPT - write me an R program ...

... to produce a **green** histogram

MA

Can you make it green

Certainly! You can change the color of the histogram by modifying the `col` parameter. Here's the modified code to create a green histogram:

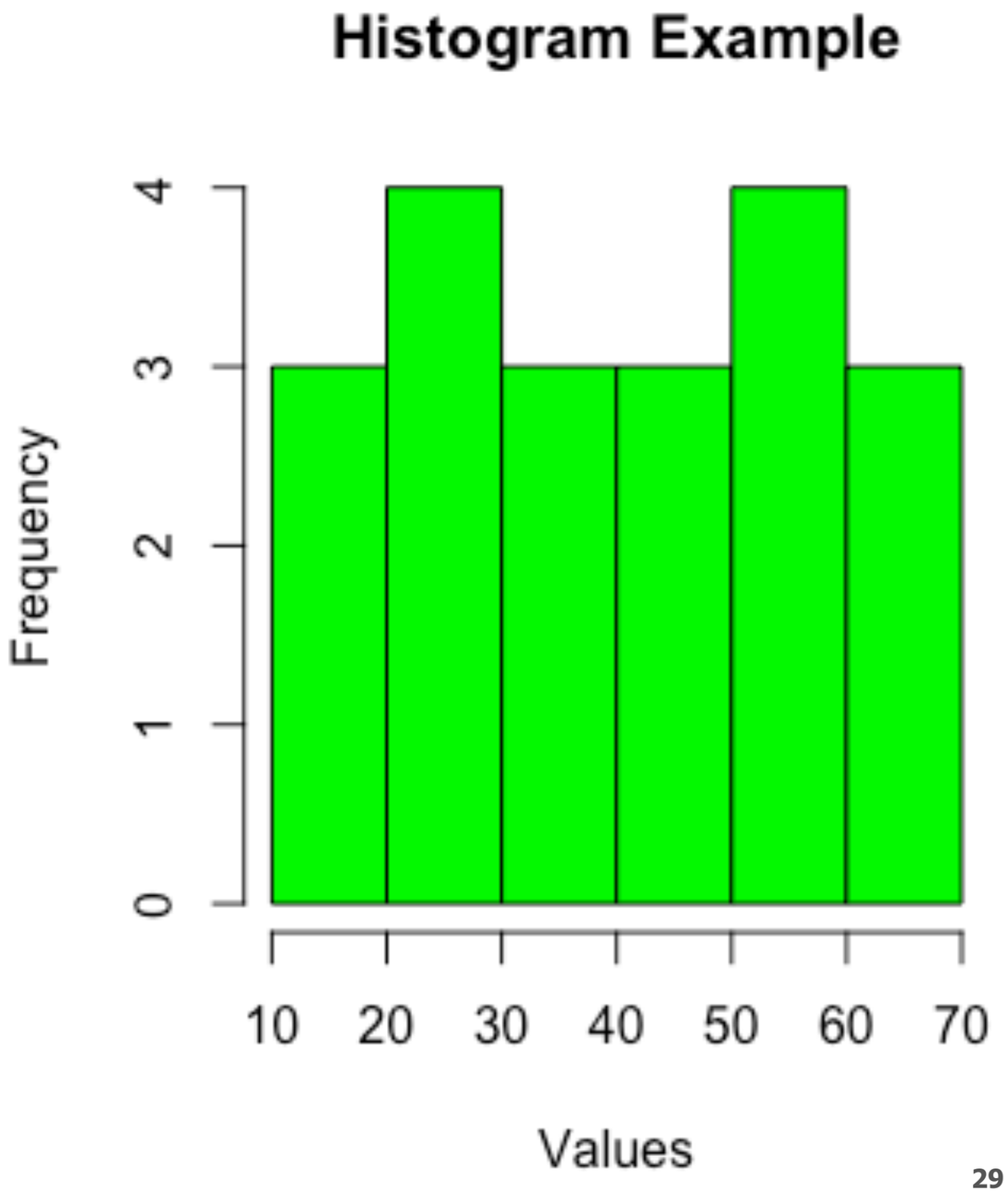
R Copy code

```r
# Sample data
data <- c(12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 5

# Create a green histogram
hist(data,
     main = "Histogram Example",
     xlab = "Values",
     ylab = "Frequency",
     col = "green",   # Change color to green
     border = "black",
     breaks = 5,
     xlim = c(10, 70)
)
```

By changing the value of the `col` parameter to `"green"`, the histogram will now be displayed in a green color. You can adjust other parameters as needed to customize the appearance of the histogram according to your preferences.

et voilà!


Histogram Example
Frequency
0
1
2
3
4
10
20
30
40
50
60
70
Values

ChatGPT - write me an R program ...

... make the data bimodal

MA

Make data bimodal

Certainly, I'll modify the example to make the data bimodal (having two distinct peaks):

R

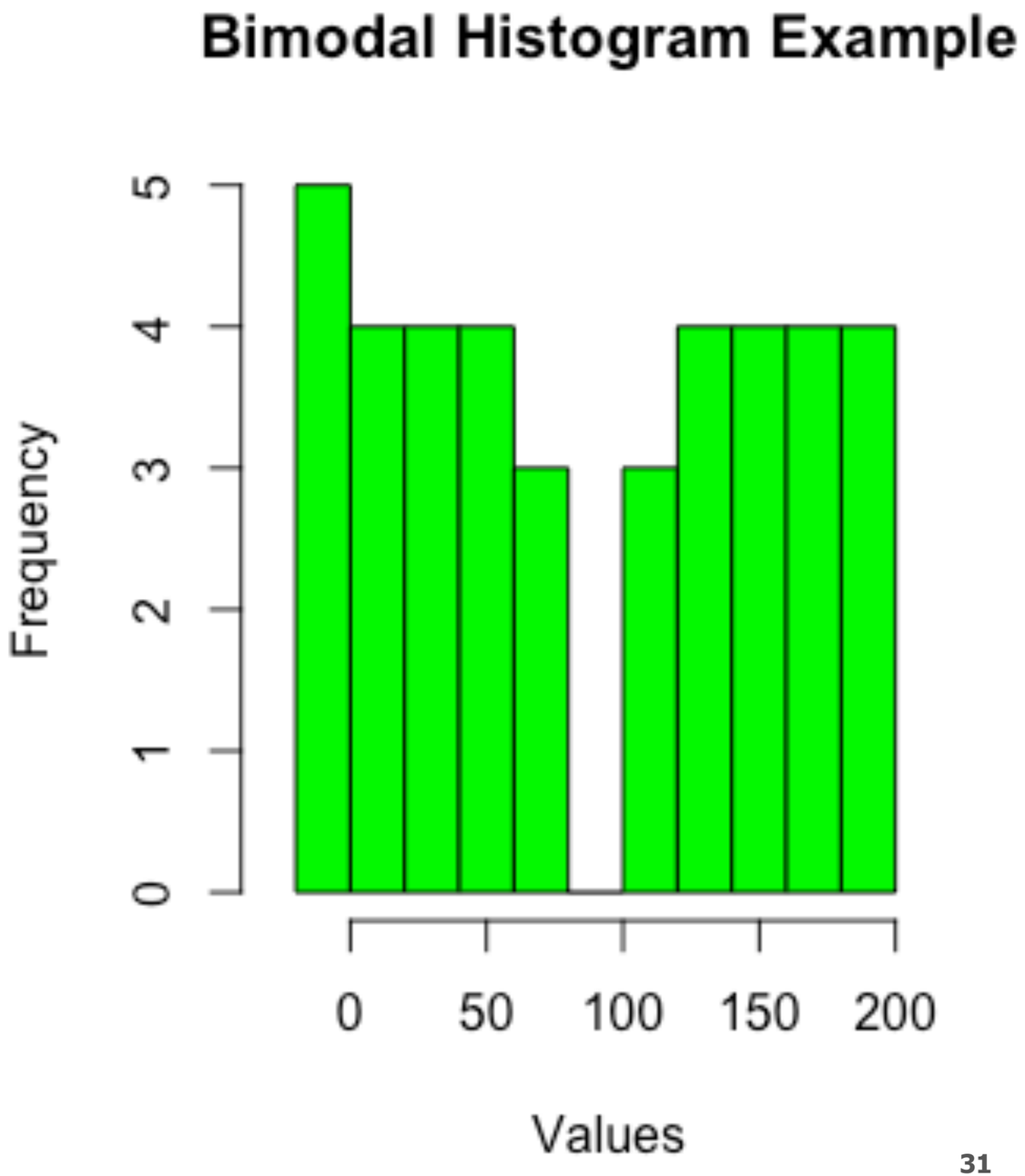
Copy code

```r
# Sample bimodal data
data <- c(-20, -15, -10, -5, 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50,
          110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 1

# Create a green histogram for bimodal data
hist(data,
     main = "Bimodal Histogram Example",
     xlab = "Values",
     ylab = "Frequency",
     col = "green",
     border = "black",
     breaks = 10,     # Adjust the number of bins
     xlim = c(-30, 220)   # Adjust the X-axis limits
)
```

In this example, I've extended the `data` vector to include two groups of data with distinct peaks, making the histogram bimodal. You can adjust the parameters as needed to suit your data and visualization preferences.

et voilà!


Bimodal Histogram Example
Frequency
0
1
2
3
4
5
0
50
100
150
200
Values

ChatGPT - write me an R program ...

... make the data normally distributed but with a negative skew

NB Turns out this was (unintentionally) a rather confusing prompt!

MA

Make data normally distributed but with a negative skew

To create data that is normally distributed but has a negative skew, you can generate a set of values that follow a normal distribution and then manipulate the data to introduce the skewness. Here's how you can do that:

R Copy code

```r
# Generate normally distributed data
set.seed(123)  # For reproducibility
n <- 1000
normal_data <- rnorm(n, mean = 50, sd = 10)

# Introduce negative skew by shifting the data to the right
skewness_factor <- 2
skewed_data <- normal_data + abs(min(normal_data)) + skewness_factor

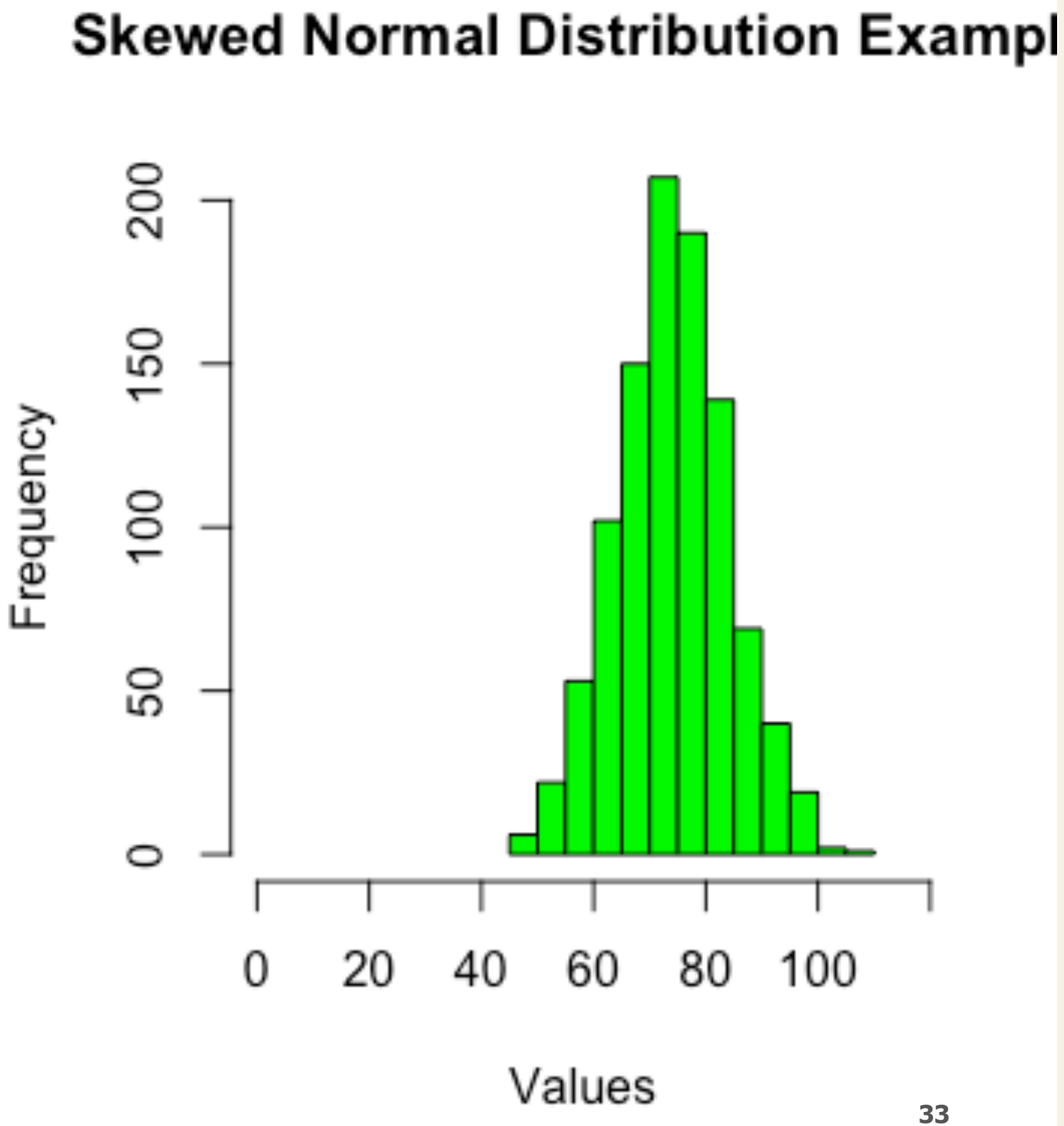
# Create a histogram for skewed data
hist(skewed_data,
     main = "Skewed Normal Distribution Example",
     xlab = "Values",
     ylab = "Frequency",
     col = "green",
     border = "black",
     breaks = 20,
     xlim = c(0, max(skewed_data) + 10)
)
```

In this example, I've generated normally distributed data using `rnorm` with a specified mean and standard deviation. Then, I've introduced negative skew by shifting the data to the right and adding a skewness factor. The resulting `skewed_data` is used to create a histogram that represents a normally distributed dataset with a negative skew. You can adjust the parameters and skewness factor to achieve the desired skewness level for your data.

et voilà!

OOPS!!!!

This isn't skewed :-(



Remember!

I had some sense of what I was looking for. I also checked afterwards to see what the skewness coefficient was.

```
> # Skewness is not available in Base R
> # so we need a special library.
> # More on this later.
>
> library(e1071)
> skewness(skewed_data)
[1] 0.065196
```

Wrong or unquestioning?

The interesting failure is therefore not merely that ChatGPT produced an unskewed result but it should have challenged my request (prompt).

Don't try to bypass learning

— Recent study from LSE[1].

— Explores using generative AI in universities.

— I **urge** you to read it.

[1] Sallai, D., et al. (2024). "Approach Generative AI Tools Proactively or Risk Bypassing the Learning Process in Higher Education", SSRN 4886015. Available from papers.ssrn.com/sol3/papers.cfm?abstract_id=4886015.

"the biggest … **challenge** of using GenAI tools … is that students may use them to **replace** their learning process and critical skills."

— Sallai et al. 2024[1]

[1] Sallai, D., et al. (2024). "Approach Generative AI Tools Proactively or Risk Bypassing the Learning Process in Higher Education", SSRN 4886015. Available from papers.ssrn.com/sol3/papers.cfm?abstract_id=4886015.

So

LLM-tools are clever (sometimes)

but they're not magic

and you still need to learn R!

5. Getting help ...

1. find/read the relevant cheatsheet
2. perspiration e.g., see this five step approach
3. talk it over with a fellow student
4. module **FAQs** on Brightspace
5. Stack Overflow
6. ask a member of the course team
7. For more suggestions visit the subsection 0.2 "vi) Learn how to get help" in the Modern Data book.

6. Extension activity

Q: Is Spotify doing data science when it recommends your next song? Is ChatGPT doing data science when it answers this question?

Read Provost and Fawcett[3] (it's only 8 pages) and see whether their view changes your answer.

[3] Provost, F., & Fawcett, T. (2013). Data science and its relationship to big data and data-driven decision making. Big Data, 1(1), 51--59. Access via Google Scholar

References

— Kabacoff, R. (2015). R in Action: Data Analysis and Graphics With R (2nd ed.). Manning Publications.

— Provost, F., & Fawcett, T. (2013). Data science and its relationship to big data and data-driven decision making. Big Data, 1(1), 51--59.

— Sallai, D., et al. (2024). "Approach Generative AI Tools Proactively or Risk Bypassing the Learning Process in Higher Education", SSRN 4886015. Available from papers.ssrn.com/sol3/papers.cfm?abstract_id=4886015.

— Wickham, H., & Grolemund, G. (2018). R for data science. O'Reilly Media, Inc.